[00:00] 发言人 1: S transpose and behavior is equal to B one and two B S. And here my question is why. H I has to be profound.

[00:18] 发言人 1: instead of another. complex function. Because we have inequality, right? H i x is equal to zero. This equality in general, and this can be rewritten as two inequalities. H i x is less than or equal to zero, and H i o x is greater than or equal to zero.

[00:46] 发言人 1: Why in my satellite those two included some pennies into you have h i is equal to zero and then you have h i of x is equal to zero negative h i of x. This is zero and since it becomes equal constraint which i must be convex then cool h i must be convex as well meaning last h i equals convex and okay there’s only one type of function outside this constraint. 啊，做饭方式。Right, that is why the inequality is a fun.

[01:28] 发言人 1: The inequality constraint is a funk. It’s a set of linear. 这是美国的规则。好的，快速的，on this site。啊，你，那就是为什么 important property physical set of complex operation problem is complex，所以为啥子?

[02:11] 发言人 1: You can prove that this set is complex. by the permission. So this is a subset.

[02:25] 发言人 1: is complex and below that the a fine a fine function will give you the hyperplane hyperplane is complex and those be calling an equality constant just the intersection of several complex set which is complex. Each inequality constraint will give you a compact set. Each equality constraint will give you a hyperplane.

[02:52] 发言人 1: The intersection of them is convex. And that is essentially to prove. Whether this optimization problem is convex by essentially minimizing a convex function over a convex set.

[03:11] 发言人 1: This essentially defines the convex optimization problem. For example, in this example, we have minimized f zero x x one squares plus x two squares. Subject to if one is devices, h one is devices, and we can prove that f zero is convex.

[03:35] 发言人 1: I compute its hessian matrix. What’s the hessian matrix of this? The Haitian is22001.

[03:49] 发言人 1: This is Haitian matrix. how how to prove this asian matrix is. positive semi definite.

[04:03] 发言人 1: improve this. 哦，you may use another tool。over at location abuse we may use another.

[04:17] 发言人 1: vector a transpose a is very important，对不对？And this is indeed true. You can just do this with application.

[04:33] 发言人 1: And the feasible set is convex because it’s defined by as an intersection hyperplane and house base. Therefore, it’s convex. And then as well optimizing the convex function over convex set. This is a complex operation for sure, but it’s not in other form because if I myself is not convex, the child is not so fine.

[05:10] 发言人 1: 啊，we can rewrite。As the convex optimization form is in couple problems other form as if zero is a convex function one is a convex function h i is the five it’s like zero and it’s zero so this is. 看我这个信息是打过。

[05:48] 发言人 1: Alright, now let’s prove local and global demand in complex of nation. The claim is every locally optimal point is optimal. And here we’re going to prove it by contradiction.

[06:05] 发言人 1: So suppose apps. is locally optimal. many other it is the mini matter over its neighborhood right, but it’s not global.

[06:23] 发言人 1: then we have some feasible why satisfying f zero y is strictly less than f zero x. This is our assumption by local optimality. There is a radius r such that every feasible c within within its neighborhood of x defined by this sphere.

[06:53] 发言人 1: we have f zero z should be greater than or equal to f zero x right. This defines the local optimality of x in its neighborhood. If their x is moist, and then we can choose this particular alpha.

[07:17] 发言人 1: and alpha is the number between zero and one. Because the other right side is minimum one are over one matter six right so it’s this time we turn now zero is one then then then is a. 康雷斯。combination of two.

[07:50] 发言人 1: feasible points. 就是 alpha is with an listener one and x one are two visible points。Since the feasible set because we have a convex problem the feasible set is convex.

[08:08] 发言人 1: And we are doing this is a compass competition. They should also intercept. They is visible.

[08:16] 发言人 1: And this facebook，and meanwhile，this is in the neighborhood of x。Right here，they are to move they matter sex，so you can start here。We move Z move an X to left hand side, this is equal to negative alpha X plus alpha Y, and we take an L to long.

[08:44] 发言人 1: on both sides which will give you. Right，which depend this remember alpha y，let’s normal one matrix alpha is this r。我可能 to this. 然后 move one alternate one minus x. 从而连环塞呃，这是 r，this means that we we have proved that this is convex combination of two visible point，this visible and they is in the neighborhood of x，and they must satisfy this equation。We have construct a particularly outside this locality.

[09:38] 发言人 1: In equality, and then by complexity we have f zero z is less or equal to one minus f zero plus one minus y, and since this is less or equal to f x, we have f zero. Z is strictly less than F zero X. Which contradictory to its local humanity right zero is this one or there is which are the zero contradictory locally local magic means x is global.

[10:20] 发言人 1: 任何商品就是 force X，就是 control。哎，刚才想的思路。Here you need to know the definition of local more local humanity and you need to construct a particularity and prove this visible in order to use this convex inequality and prove this in the neighborhood of x to using this locality inequality and this two inequality computing between each other.

[10:57] 发言人 1: I can say this is global. 呃，Reality。we have standard form problem not necessarily convex minimized over f zero of x subject to h f i x equals zero h i is equal zero here not necessarily convex means that f zero and f i.

[11:30] 发言人 1: are not necessarily complex function h i are not necessarily profile we come from convert this customer relationship function into its lagrangian form. which is R N times R R times R S is maps bar with the domain is the. Product of this with that, and here we have the branching L as a function of X input, and we have two sets of branching multipliers lambda, and the new lambda is associated with inequalities, new is associated in inequalities.

[12:20] 发言人 1: It’s equal to f zero x plus the sum of i from one to r come by five of x plus i equal one s divide h i of x. So the algorithm is basically the weighted sum. Objective and constraint functions or linear weighted sum，and it’s a linear combination，and lambda i is an approaching multiplier associated with f i of x。咳，类似 zero，UI is not launch much，associated with China。I remember to build this approach, you have to work with standard phone, so the equality should be always be this hour with you.

[13:18] 发言人 1: 所以，If it’s written on the two，and minus sign reverse order，if there’s some now negative term on the right hand side，move it to the left hand side。Make it zero all right similar for the H O X. May from the launching we can compute.

[13:42] 发言人 1: They are branch due function. where it is g of lambda u and this is equal to the informal you can. You can hear, understand, inform as just.

[14:00] 发言人 1: Minimum informal x is in the set x l x lambda u where we treat lambda u as constant. 啊。We know the definition now, mounting is just a minor with the competition of this. And here we can prove that G is concave. G is complete and can be negative infinite for some lambda and u. because we’re doing the minimization and if we can and this operation problem minimization problem can be undefined on below so it can be that thing right but in general g is concave so why g is concave. So why she is computer for all possible functions?

[15:23] 发言人 1: So even let’s say even a face x, so what inside the parentheses even a face x. Even if it says the ranking is just a profile function respect w right. because h i x is just a coefficient multiplied by h i is another coefficient, so it’s linear, it’s linear, that is. 是。

[16:05] 发言人 1: 咳，你好。That is, let’s say if we draw a figure like this, the x axis is just the range multipliers, and the y axis is l x lambda new, right? In terms of lambda b, this is a fine, right?

[16:28] 发言人 1: This is a function for let’s say a particular x x x zero. This is another function for x one. There is a third function for x two. And here we are only we asked to take minimum one. Right，the minimum one，that is this part。

[16:58] 发言人 1: And that’s why it’s come here. all right. And we have a lower bound property, if we constraints lambda is greater than or equal to0, we have the not branching dual function for lambda view is always going to lower bound the optimal one piece up, and the proof is very simple, so if.

[17:32] 发言人 1: X is X is feasible and lambda is greater than zero here 呢? If the f hand for this in funny symbol great equal to if the if inside the matrix. It means that we are dealing with all the heavy all the same left address if they left is a vector it means that.

[18:00] 发言人 1: Each entry should be greater than zero for all entries. Then f zero x theta should be greater than or equal to l x theta lambda 六。So how to justify this inequality? This because x theta is feasible here we have.

[18:27] 发言人 1: Definition of L recognizes F zero x theta h i theta h i theta with those at h i x theta is zero and this is less than zero right lambda is greater than zero as specified and the second term is less than zero. That’s why it’s zero it’s delta is equal to l delta and to and we minimize we minimize over the informal. And which just why the second quality and the informal over x gives you the number running and this is for for the left hand side this is for all feasible exterior and we can minimize. Overall feasible gives the primal gives the primal optimal value of p star p star is the equation of optimal w not.

[19:43] 发言人 1: So for any visible point you should be greater than because the inequality is. 啊，这个 miniimize over the. Any feasible points is our bound for the upper bound function。I will minimize our overall feasible points give the which is also we drive the g number six。Let’s do some example, this normal solution for linear equations.

[20:21] 发言人 1: We minimize x x transpose x subject x is equal to b. The due function is x transpose x new transpose a x minus b. Right, this is not in standard form.

[20:39] 发言人 1: We must write something like this: x minus b. And here this is system equations right and we need to add the vector the number not much vector and transpose it as an inner product right this the. Can be written as an inner product.

[21:03] 发言人 1: And to minimize l over x, we set the gradient equal to zero. And the gradient of l respect x is equal to two x. And transports new is zero zero and we meeting mass.

[21:18] 发言人 1: We saw for x this is equal to negative one to a transpose u and we plug x into the original formula we plug this into the launching this gives you negative one or four new transposes a a transposes new minus b transposes new which is a concave function. Thank you. And lower bound property is done.

[21:46] 发言人 1: It’s always gonna be lower bound by this for all new. Are they so a larger larger. Found this more useful route on right and then we can find the best view by math in the right hand side in order to find it.

[22:15] 发言人 1: And why this is a concrete function can improve it. So this is concave. meaning that one over four new transpose a transpose.

[22:33] 发言人 1: you. Class we transports new this is complex. Right, and then we can compute this hessian. The hessian matrix is simply one over two a a transpose. This is the Haitian matrix.

[22:58] 发言人 1: And ready to computer. We need to prove this H matrix is positive definite right by definition. This is equal to one over two X transpose A we work with for any X A transpose X.

[23:18] 发言人 1: And this is equal to one over two, a transpose x transpose a transpose x, and we can delete. A transpose x as another vector that this b transpose b which is equal to. So we define define a transport sex equal to b this is going over to the sum b j squares which is greater than zero.

[23:48] 发言人 1: They will prove this metric h metric is positive semi definite and we prove it. This function is convex and i am on the side. comes country that’s how you prove this equation this country and for now want to do functions always gonna be country.

[24:11] 发言人 1: And have started form in your. Programming we minimize the inner functions three transpose x subject to linear constraint a x is equal to b where four x are non negative then we have to function the branching l and w is equal to c transpose. And we have this convert to the standard form.

[24:42] 发言人 1: We have new transport a minus d minus lambda transport x, and we can combine common terms which gives you that to be transport new and a transport new minus lambda plus c transport x. And l is a fine in x if we minimize l. It’s gonna be generally put negative infinite, right? Unless this vector is equal to zero.

[25:14] 发言人 1: If it’s equal zero, the infimum is negative B transpose new. Otherwise, it’s going to be negative infinite. therefore and we get rid of this that to infinite as a low run. the is no for everything which is useless and they need to have a finite lower right therefore g is linear lambda.

[25:43] 发言人 1: Uh，which is negative B transports new，and we have a constant a transports new matters，and the passage equals zero，and then we have a lower bound property，piece done is where can I go to negative B transports new if. This is positive and here we know that a transport new are six greater than lambda and lambda is associate with this inner. inequality and it’s constrained to be positive and we don’t constrain the new associate with equality we only constraints that are run to market associate with the inequalities right.

[26:31] 发言人 1: And then we have due problem not launching due function due problem we maximize respect to the due variables lambda new gene lambda new subject to lambda is greater than zero and lambda new are still feasible if lambda. is great great is now negative and now we are listening to dream and we are fans of best lower bound of him and. This is a convex optimization problem, and optimal value deduced by this time.

[27:06] 发言人 1: And we have weak duality, meaning that always holds for convex, nonconvex problems. It can be used to find nontrivial lower bound for difficult problems. And this is very useful for a very difficult.

[27:26] 发言人 1: So let’s say we minimizing over F L X subject x is in the set. This is non convex, right? 意思。And then we work is we computer is not running identify is not want to do function.

[27:52] 发言人 1: And optimize the linear programming problem compute d star and without d star for this problem this is equal to five. And even though we don’t know the optimize, we may have some feasible point, right? This is feasible. This is we can computers.

[28:22] 发言人 1: This the value of this right，and which is turn out to be five point one，and you know that this exam is close to the optimal because it’s very close to this around。Right，so this is why the low fund is very highly until low fund is very friendly，because you can do visibility test，test how close this final point close to low fund，and if it’s close now。啊，喂喂喂，三喂，我是我是。

[29:00] 发言人 1: That’s that we can stop the population, and then we have strong duality. This time is equal, this time and for non convex problems strong duality holds on the shoot for regularity conditions such as data condition, data condition that so there is a point that is strictly feasible. strictly zero h i o x equals zero.

[29:31] 发言人 1: Making more safer conditions. So what is the animal? And then we have what’s known as commentary scientists.

[29:41] 发言人 1: So assume strong duality holds. X star is prime of optimal lambda. Bill is still optimal.

[29:50] 发言人 1: 这是去 star。This is D star and D star is computed from the informal of the branching. This is the branching x. Lambda star new star and this is equal to f zero of x star remember we only know that x star is a minimizer of f zero it’s not necessarily a minimizer of this launching l.

[30:24] 发言人 1: So this is less than or equal to and since this is less than or equal to zero and this is less than or equal to f zero of x star and you have f zero of x star those are the two endpoints therefore the two inequalities hold with equality meaning that. X is not only minimized at zero of X but also minimized this launching where we happy not a star this now and this is exactly zero. Now the I if I start is equal zero for I from one to r that is complementary statistics.

[31:09] 发言人 1: For example, if not the I sweeping for the people. 那有 F I X is equal zero，if F I is strict。negative lambda L star is equal to zero.

[31:24] 发言人 1: So this is comprehensive strategies. And then we have, we summarize those into KKT condition, crush, common, Tucker conditions, the falling for conditions, for KKT conditions for problems to match by five. Versus primal physical visibility, we have H1 bases, this and zero, H1 to zero.

[31:50] 发言人 1: 啊，那些。Primary feasibility we have due feasibility not nice we we have zero we have components like this which is encoded here. Now the I F is equal to zero.

[32:09] 发言人 1: We have gradient of black range in respect to explanations that encodes this information minimizer. So the gradient of the zeros. Plus the sum of lambda i gradient of f i plus new the gradient of h i is zero and if strong validity holds.

[32:33] 发言人 1: And X，那那六，ultimate must decide，I give you conditions。是 strong reality. 看。Derive taking the condition and not vice versa. 价格明显。But we do have sufficiency so we can we we we can, 对吧?

[33:00] 发言人 1: vice versa for convex optimization problems so assume f zero of the far achievable and convex on the convex domain x h i s are fine if x not that new satisfies this k k d condition it’s primal due optimal. And the proof is also very simple. The things number star is greater than and now launching.

[33:30] 发言人 1: is convex. And the grounding is complex in x then stationarity. If G by definition is defined by informal of X, and this is equal to. this one because if it satisfy the k k d condition, it’s also the minimum of this right.

[34:01] 发言人 1: And we have primal feasibility and company microscientists gives you this since they complementary scientists give you the second term is equal to zero and this is equal to this and by weak duality g is equal to this this side this side this side this side this. And then we have. The endpoints are equal or equality equality 啊，both travel until. Alright now let’s take a ten minute break we will continue at the server 我已经。

[34:52] 发言人 1: うん。

[36:57] 发言人 1: 成绩加入成绩。

[37:19] 发言人 1: 成都的。

[38:03] 发言人 1: 嗯。

[38:56] 发言人 1: 好的。

[40:18] 发言人 1: 笑死。

[40:27] 发言人 1: 可以关注。

[41:35] 发言人 1: 对，然后你就放个壳，你们就没整，就放个壳在那拍照。

[41:45] 发言人 1: 我。

[42:30] 发言人 1: 但是使用同一个数据库。

[43:17] 发言人 1: 上面的记录。

[43:30] 发言人 1: 好家伙。Alright，let’s continue。And the first lecture maybe abstract to you. And in this lecture, I give you a company example, how to from you to from travel to do and get my one into function.

[44:04] 发言人 1: And the inner classifiers, we have many inner classifiers, separate class using hyperplane, and example inner discriminant analysis or LDA, logistic regression, support vector machine, and here is the machine with primal form soft marching version. The minimize respect w b and c one virtual normal squares of w plus c. 他们是上 iphone one to m c i more。One I W transport some speech we great one minus C I and such in our entity which one which I.

[44:54] 发言人 1: And what if the data is separable but not linearly separable? As show in this three cases right so we can easily draw in this three cases we can easily draw this is hungry. 啊。right.

[45:21] 发言人 1: To class distinguished cases. 打。机场。The idea is to transform the input space map adds to a new high dimension space.

[45:40] 发言人 1: From N to L，typically L is greater。感恩。And then we learn in the customer in a new space. And in this case here we have. In this case, we have n is equal to.

[46:08] 发言人 1: In this we have n is equal to two, l is equal to one, from x one we create another feature x one two, and we. 你的 class 摆一起，for this n is equal to two, l is equal to。你呢？N is equal to one N is equal to one only one feature.

[46:37] 发言人 1: And it’s equal to one, where l is equal to two, like what two in the class, in general. We want to leave today feature into a higher dimensions for inner classification. So let’s try it two dimensional vector include x one x two with transmission consist of all these coordinates terms that transfers. and. Square two times two is two squares, and then.

[47:13] 发言人 1: So they when then we want as them on the transform space the decision hungry the margin will be straight line or have a plane in the transform space when we map the decision hungry from the transform space after a war space input space the decision hungry is going to be nothing as shown the figure. So SVM with transform input to illustrate the feature mapping. We first consider a hard margin SVM, assume the transform data is intersectable.

[47:56] 发言人 1: And here we optimize WB one. Over the home squares w subject to its functional margin should be greater than equals one for all training examples, and we apply high dimensional transform to the inputs。And the only difference is replace.

[48:24] 发言人 1: X 三。With fine，our fine，you are fine，is a very high，really fast。And now the long vector is using the l dimensional space if l is very large and you can see in the later size l can be.

[48:55] 发言人 1: 但是下载用户。And calculate the vector for the critical point, we have machine or even increase the beyond because f l e c imposes the infinite dimensions. And optimization be very efficient to have space，然后。So here we’re going to assume from primal to deal, we’re going to derive this.

[49:39] 发言人 1: So here we first write on the. 没有。从。哦。Soft module understand with minimize respect to a big vector astronaut psychic marvel. 啊，over to W transports W 啊，C 三，I. Subject to Y I W transpose X I plus B square one C I for I which is one after PM.

[50:32] 发言人 1: 啊，这是 iPhone so market。And we can write it in we can write it in standard form. This is not standard form, right? We can write it in standard form. That is negative.

[50:53] 发言人 1: Y I W transpose X I plus B minus one minus C transpose C I。嗯，是他们去零，他们还不太清楚，就是 I is in some zero。来，this is three four。And then reform the project.

[51:26] 发言人 1: 嗯，那不让走，那也是也要。好，W W B C 阿尔法贝塔阿尔法 is for the first equality 贝塔 is for the second equality。That is equal to one over W transpose W plus C the sum I from one M to C I。

[51:55] 发言人 1: 然后，manners。这。Sum apple one to m alpha i y i after transpose x i plus b minus one plus c i of minus the sum apple one. 不能被他爱，说哎，this is an advantage，so this time.

[52:39] 发言人 1: So that is one over two w transpose w minus this is quadratic term respect w and have linear terms respect w.

[53:02] 发言人 1: 嗯。啊，那也不好，那上 iPhone Y，M R，华 I，Y I。B plus some iPhone one，two C I C minus alpha I minus beta I，and we have some iPhone one today。马上报案。

[53:35] 发言人 1: 喂，this is. This is the term respect to w. This is the term respect to b. This is the term respect to c. This is term that is constant.

[54:02] 发言人 1: Right. And now we want to after this the launch we want to minimize the launch with them to come variables as that’s out of me. To do that we take the gradient of this air respect to w which will give you.

[54:25] 发言人 1: I feel the first two terms minus the sum iphone one to M alpha I one I. X I V C to zero。啊，which have to be about the same term，啊，这是 negative 的。

[54:49] 发言人 1: iPhone one M 阿尔法 I one I is equal to zero，and we can get rid of this，stay to sign。And we have L with the C is equal to C. 可 see i respect 可 see i this is to see minus 阿尔法 i minus 贝塔 i is to still or i see for one up there.

[55:26] 发言人 1: right.

[55:36] 发言人 1: 安城。Now we plant. Now we write this explicit for this we have w is equal to the sum linear calculation.

[55:56] 发言人 1: of its input vectors. And by plug this. into this formula, this disappears. 啊，THIS.

[56:12] 发言人 1: Here this appear disappears and we then we only have this customer remains and we have this term where we put w. Size now, let’s continue our derivation. Therefore we have. 及那忘记丢方讯及。

[56:43] 发言人 1: 啊，G 哦，阿尔法是一个通话 over two，W 他说 W，over W is a good two 啊。啊，妨碍。Why I X I translates here we use another.

[57:00] 发言人 1: exchange is one of the alpha j y j x j. And with Mars the sun iPhone one 苏联 iPhone I Y I. This this is W this is some J one and alpha J one J.

[57:31] 发言人 1: X J，X I，constant。And we can write constant funds. 啊，for the first term，this is equal one over m，the double sum，i find from m。接通关 M 然后 Y I Y.

[58:00] 发言人 1: J 阿尔法 I 阿尔法 J X I transpose X J。minus double sum。Y I Y J 阿尔法 I 阿尔法 J。X I transpose，let’s see，and we have the common terms。And we will give you the sum alpha one to them alpha i minus.

[58:46] 发言人 1: One over two，我上来从 one to m，直接从 one to m。Y I Y J 阿尔法 I 阿尔法 J 方 X. 阿尔法 I 阿尔法 J。X I transports X J. Right, this is for the money, you know.

[59:32] 发言人 1: 好，我们揣一次。

[59:41] 发言人 1: 然后回来，然后往对方群，他自己要装。So here this is i from one to the alpha i minus one over two the sum i from one to the end the sum j from one to the end. 晚安，YJ 阿尔法，I 阿尔法，J X I，transport exchange。啊，you want maximize, i want to do function，this is not want to do function，or i want to do problem，want maximize this function。It is subject to to have constraints, this this this constraint and this constraint and the primary durability constraint.

[60:31] 发言人 1: Subject the sum of one to m alpha i one i is equal to zero alpha c minus alpha i minus alpha i is equal to zero we have alpha is equal to zero beta is equal to zero this is. 有这个能力，然后 this this is exactly equal to. 好好刷。あ、まさに。

[61:05] 发言人 1: 阿尔法 X 为零，and 阿尔法。I is this i could see right that is our eyes we like zero this i could see. 啊，这是他顺。啊，对，you can remove this card。And you can remove this part and right the same in quality as alpha I we can put the reason to see for i see one after and once on ice determined by to compute better i see minus alpha.

[61:49] 发言人 1: Right, so here this platform and this is already. Develop for you, we first convert this in standard form and write an approach, and we take the derivative of L respect to W, B and C, and set them to zero, and put them back to the approach, give you this form, and have run. And we put this together with a constraint and eliminate beta right because c minus argument equals zero and we arrive at this formula as we.

[62:37] 发言人 1: Show here where. So this is our sun of our one and our two the sun double sun. Here this is double sun. X 三 Y I R Y. Alpha j x i transpose x j subject to the sum i alpha y zero zero where alpha i is we like zero example c alpha i corresponds to the ice training example x i and we can recover w as the sum of the training points. I have some reports.

[63:28] 发言人 1: And when we identify when we solve this new problem we solve for alpha and we can calculate w we solve for w and later i will give you the details of the movie and they will recover the function we can classify the points with sign. So interpretation. The interpretation of alpha the combine complementary.

[64:00] 发言人 1: Likewise，with primal feasibility，due feasibility，we can show that if alpha i is equal to zero，we have functional margin is with that y，it’s on or above the margin，as coefficients in the decision function zero。is zero coefficient is decision function is zero much. If alpha is strictly greater than zero less than c we have functional margins that one if the exact margin and if alpha i is equal to c we have this is this one on the margin or values margin.

[64:46] 发言人 1: And for this part this gives you the calculation formula for for e right so after the job position is solved the optimal question can calculate you than the training example. 博士。where 阿尔法 is strictly 大于 zero 之类的东西，for example。WE HAVE Y I. W 三。As a new weighted combination of your examples.

[65:23] 发言人 1: X plus B star is equal to one on the function margin, right? We only use points where alpha is strictly greater than zero, this and c, and it’s exactly on the margin, and that we can work with a stress one. We can compute y squares varies w star transports is plus.

[65:49] 发言人 1: Besides call to one and why squares is equal to one because why is either one negative one and they have. W star transpose x plus b star is equal to y and b star is equal to one minus w star transpose x and you are able to b star and then if you have multiple points right if you have multiple points. Does science by this included in the set you you can complete these as the category of this you the sum of.

[66:35] 发言人 1: The sun always why is this said to you you have. One minus W start average of this option. Right, so I will, I don’t have time for the detail duration for this three cases, but I will put it in in the lecture, so that you guys can take a look at it.

[67:09] 发言人 1: And then here shows this example where alpha should always not be positive here. This should be positive. And we have non linear decision boundary.

[67:25] 发言人 1: we have alpha is equal to zero and alpha is equal to three zero. And then. So the for many for the twenty sixth the majority of examples is beyond the function one is beyond the party decision.

[67:50] 发言人 1: Margin around when alpha is zero right when alpha is zero they don’t contribute to the normal vector. The only input samples that contribute to the normal vector is those samples with nothing or. 哎，it is so simple with some.

[68:20] 发言人 1: And the supported actors, the supported actors has a precise abstract definition where they are not not there, and they contribute to the calculation of the Nobel, they contribute the calculation of the Nobel actors. 好吧，啊，this is very efficient computer，maybe。Among this tens of thousands only a few twenty and zeros and the very smart sky bridges.

[69:11] 发言人 1: So kernel，the kernel function，the US one。you form as transform inputs right here we remember we this is form. In the product we transform it.

[69:31] 发言人 1: 就是丢还是什么？With transform input，we replace x i x j with five x f x j，and it can complete reading in terms of inner product。So rather than explicit calculate the high dimensional vector five of which five.

[69:56] 发言人 1: We only need to catch the inner product between the two vectors. That’s why we introduce a curl function. A curl function takes two data points as input.

[70:07] 发言人 1: And is the inner product of two data points in the transport domain. It’s much less expensive to compute than explicit calculate a high dimensional fish vector and the inner product. Here we show this polynomial kernel. well. the input vector is.

[70:38] 发言人 1: X one up to X three is in the R N。And kernel between two vectors is the p polynomial or polynomial where the kernel xx prime is equal to x transpose x prime to the power of p as it’s equal to some iphone one to nxi. I prime the report.

[71:03] 发言人 1: 过。And then from this we need to figure out what, what’s the corresponding transformation, right? What’s a cross talking transformation? And this is equal to the sum of I from one to J, one to N, J from one to N, X sine X sine prime, X sine X sine prime, and this is equal to five X transforms five X prime.

[71:35] 发言人 1: And the transform switch is just a quadratic function of input x one x one x two up to x n. And you can compare same number modifications for kernel. We only do an inner product.

[71:53] 发言人 1: We only do this inner product, right? which is under order of. And if we do this transformation, we first construct this feature transform, which is on the model and squares, and which we do the inner product over the transforming still on the model and squares.

[72:18] 发言人 1: As you see, you have a company pressing up like product introducing the color function. You reduce the competition costs from big O and squares to big O one. 咳呀。呃，current tree 作为 place in the product。With the kernel function in optimization is called a kernel trade.

[72:45] 发言人 1: It turns the inner function from another one and the shape of decision boundary determined by the kernel. And in this case, we have kernel as well where. 这是 constant term.

[73:00] 发言人 1: Is is the same minus one over two the sun remember still double i j from one to m y i y j alpha alpha j here originally we have inner product right originally we have inner product here. And then we move to the transform space. Then we replace the inner product transport space with a kernel function and subject the same equality and equality constraints.

[73:39] 发言人 1: 啊，production can also be computer using the current function。Because still in a product between ice example and the cream example. All right, and invite lecturers will say color tricks。The other machinery are such as the linear regression.

[74:06] 发言人 1: We we can use turtle the linear regression. Or we can talk about principal component analysis of PC，we have a kernel last version called kernel PC, which exactly using the same trick. So the only thing you do is to identify whether there are inner products between two eight points，if we have.

[74:32] 发言人 1: The new product in the formation you can readily replace it with a kernel function and we have different kernel function work with which will give you different inside the others. And this is the kernel as one with polynomial kernel. The dissenting surface is a cut of a polynomial surface.

[74:58] 发言人 1: So higher polynomial order uses more complex decision boundaries. And we also have RBF kernel. RBF is short for radius basis function, which is very close to the Gaussian formula, but into the gamma of squares of a. takes six prime because the doctrine the expert also protected terms in the data points and gamma is the universe time this time of the current controls the smoothness of the country with a small gamma we have wide doctrine which needs to smooth function without come to the same doctrine which need to weekly. 啊，第二。

[75:56] 发言人 1: So that we can identify the corresponding feature mapping.

[76:09] 发言人 1: And this feature mapping is belongs to the positive infinite. 可以出 I P 还是 influence 是外一张这是 body. Because in the derivation of this you will need to use the exponential function let’ s say of x and you need to do the Taylor expansion.

[76:49] 发言人 1: 嗯他真的。他心里有你我我重。不行。And this gives you all the information. And you can try different gamma, each year, different most of the Hong Kong, the situation, and we can do the kernel.

[77:27] 发言人 1: Okay，kernel，that’ s the summary。So kernel classifier defines the shape of the number，produce boundary，increases transform the input vector for high dimensional space，may be infinite dimensional，and we use it in the classifier high in space。And the same hour is not in your own to this training.

[77:51] 发言人 1: We balance the large march against penalties for large validations. The margins better the transport bigger space. Seek control.

[78:00] 发言人 1: The trajectory of numbers that you are inspiration was strongly we use hospitality to see a permanent article at here. For curl as well，we introduce to label of hyper，oh，original label of hyper，my switch produces。I have different classes.

[78:24] 发言人 1: I think this in the class we we just you two okay. One is polynomial curl, the other discussion curl. Which one to choose? That’ s why I have a problem.

[78:36] 发言人 1: After you select particular color for them. How to set the bandwidth the gamma is another parameter. So the advantage. Of curl SPM tsunami，就是那个什么，就是复杂的，一个方法，第三个，distance to the curl function。Rules such as they are from other factors computation expense across vibration and you calculate the curl matrix which is remember curl takes two data points as input therefore the little compute and square terms where m is the size of twenty five for large m.

[79:25] 发言人 1: 有审核，我拿去吧，统计的时候。And another, so if you put all color matrix, it’ s a color matrix. Color matrix is magic.

[79:43] 发言人 1: And sequential minimum oxygenation，that’ s a method for me as well。So before that，we first introduce coding distance，to consider the unconstrained optimization problem，we minimize respect。Theta，还有系统，把系统啊，Theta 真的。

[80:04] 发言人 1: The coded distance，that’ s the first selects initial point zero，as in the great distance and repeats。Here we want to optimize the whole system，you only choose one direction，and then。And then we optimize only this parameter while holding the other parameter fixed.

[80:30] 发言人 1: At each situation the algorithm determines a coordinate and minimums over the corresponding direction while fixing all other components. A coordinate is a design for both dimensional and non dimensional topics. Why not the people come course coordinating this is how work is very little dimension dimension operation with a one or two in one or two 然后我们。

[81:00] 发言人 1: Use great information we can use another zero orders information for nation, and that’s why it’s differently for now. Potential context, and here choosing a court reason on for that function, we first opened on one axis get here, but on x axis or present on one y axis exactly y axis. 啊，那个就是这个 directory。

[81:34] 发言人 1: And S M O or sequential minimum position is indeed a coded decent position method. As you know, is an algorithm for solving the quadratic programming problem that arises during the training of SVM. You might like John. I might if I remember right, I might have personal research. It’s frequental, and it’s not parallel.

[82:00] 发言人 1: But only all times a set of tuna lunch on times minimum but all times small is possible some problem at least that relation decides whether constrain for the children care of not one my parents is consuming occupation. recall the dual form of carbonize as well as soft margin at about going to not like this occupation problem. The S M O decomposes a problem into its extreme.

[82:38] 发言人 1: optimizing a minimum subset of just two variables. 所需要的什么? Ask why。出一块儿，我是多图案。你。

[83:08] 发言人 1: So the smallest support should be only one purity right so right in this case the smallest possible support is. involves two developers. This is because we have this linear equality constant, right? If only just one the equally consumer will be violated.

[83:39] 发言人 1: you need to adjust one and select another one composite point in order to respect this linear equation constraint. That’s why the minimum of service for you. Two variables is for this equation.

[84:05] 发言人 1: So each two variables are program can be solved analytically. Why it can be solved analytically? Because it corresponds to one dimensional correct function. We know the formula. Right to show the one that you know that happens.

[84:29] 发言人 1: And the overall arguments remains to it right after solving this pair we identify another pair and wait it was identity another pair and it is this condition this linear. This equality constraint is enforce through the out the tradition impact of the smallest number of math are built at each traditions two whenever one math parents updated at least one other. Other matter need to be adjusting or to cyclists condition.

[85:05] 发言人 1: At each at each step, a simple chosen of the two develop alpha alpha j given not for the other subjects. And the choice of two derivables is determined by heuristics, which will all be covered in the size. But we will try to implement it in the iPad.

[85:29] 发言人 1: Local detector for considered. For example, the violation, how. To what extent violation the community conditions and it’s the inner loop population the operation to develop small inexpensive after to make it’s enough efficient and performance of course depends on the data curl stocking. So now let’s do a classification inside.

[86:04] 发言人 1: The classification task，so observation x typically a real vector of feature values exits in dimensional space，and classify from a set of possible classes, why this for value cases positive one left one，and why put the left one。But we want to delete the effective classes, take one because it’s either for loss components. Right, we can compares L zero one loss, which response classification errors.

[86:40] 发言人 1: We want to compare. majestic loss，right? we also want comparison loss。嗯，还是说 as exponential or as point zero one。And for k nearest neighbor, the type is discriminative class, cover this multiclass classification distance from this neighborhood.

[87:09] 发言人 1: There’s no training needed. As long as you have sufficient examples where k is equal to one, you always know that. It has infinite capacity to approximate any decision factory for care.

[87:26] 发言人 1: Well, general customer have this generative class much class. This is about the genuine nonlinear training estimate class condition probability of this. You require by maximum like with data, and you will always have to estimate she or why. S class.

[87:46] 发言人 1: 出来。And the classing summary not just regression discriminative binary is in power inner remarks like you put in shop by five and. Because only loss is modest loss.

[88:05] 发言人 1: Well, machine discriminatory battery slender. Well, penalizing violations. This is for. The underlying losses hinge loss right for logistic because it’s not one us equal next three for hinge losses marks them zero one matter see we have curls of the big machine type discriminative battery.

[88:35] 发言人 1: Now inner and maximum margin in feature space of opinions like accumulation in inner for what’s this up the inner or set variable violations of margin and put it in object as some models have terms to cover the over thing of trend data this can improve generalization to do data. 啊，有时候他出了 controller。The visualization that exists on the training or validation set we need to help out the announcing data set for testing.

[89:12] 发言人 1: And others in smart class application can use better class to do multiclass using versus rest population and feature population normalize each feature dimension so that some feature dimension with not arranged small dominant population so data balance if more data in my class then I wait to each class. To and the objectives. And class imbalance mistakes on some classes are more critical.

[89:42] 发言人 1: You can bring ways class focus on class for on predicting predict one class at expense of others. And other things applications we have a lot of applications because we have got. It’s not getting face gender regulation, face detection, digital classification, and features choice features is important using only my features may confuse the class bar using domain knowledge to express face which from data all this right we.

[90:21] 发言人 1: Don’t do had engineering or feature design with you. You learn to learn the feature presentation as by the decision boundary jointly in entertained fashion. Which class by expressed there is no free there there is no free lunch

theorem in in machinery under uniform average over all possible top countries no learning algorithm has an overall advantage use learning useful learning therefore depends on the assumption of the problem cover.

[90:57] 发言人 1: In other words，there’s no best class or tasks。The best class fire depends on the particular program，and this data sets as you can see。And we do have a support machine, the mini batch version, so that’s great condition to implement for you.

[91:20] 发言人 1: So I encourage us to take a look at it. So if I remember correctly, we also have some optional material, right? So the SVM of simplified version of S and O algorithms in this for you.

[91:44] 发言人 1: So you guys can work with those analysts to our side of the code. Is measure how to select those two you like what’s the trick what’s the risk. Or adopt or implement in this particular ipads notebook and how to implement those. Alright, so that’s all. For the lectures，I will come take a break，I will put enter portal for today section。

[93:55] 发言人 1: 这个这不是你的讲稿吗。你。我的意思是要不要把成品给它缝上去，两个看一下。

[94:18] 发言人 1: 不然他们以为我们没做人了。

[94:40] 发言人 1: 我还可以画那个技术图嗯。因为毕竟从外面请了专家嘛。

[95:47] 发言人 1: 呃，操作。

[97:22] 发言人 1: 那怎么。一起。

[97:32] 发言人 1: 现在看。

[98:21] 发言人 1: 工程。

[98:42] 发言人 1: 我。请。